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Service Description (SD) Arrowhead Z-wave System Demonstrator – SetPoint Thermo service

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Abstract

This document defines the template for the Service Description of Arrowhead compliant Services. A Service Description provides an abstract description of what is needed for systems/devices/units/software to provide and/or consume a specific service. SD's for Application Services are created (specified) by the Pilots WP's and by the common Arrowhead framework. The SD shall make it possible for an engineer with technical programming knowledge to achieve an Arrowhead compliant realization of a provider and/or consumer of description of how the service is implemented/realized by using the Communication Profile and the chosen technologies. A Service Description (SD) is the Service in a specific technology. All systems/devices/units/software implementing an Interface Design Description which complies with this SD will be able to exchange information with each other. All Arrowhead Service Description should be specified using this template and stored on a common repository (available on the SVN server), in order to document and formalize the pilot demonstrators and the common Arrowhead framework.

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1. Service Description Overview

This service Add current SetPoint Value from the Thermostat into the Thermostat Device manager.

The encrypted JSON object contains the following fields:

Request:

```
  {
"serviceType" : "SetPoint",
 "deviceType" : "Thermostat"
  }
```

Response:

SetPoint Value is added to the thermostat device manager with unique ID and type and the time stampand the SetPoint value is returned.

```
  {
“setpoint” : Value
}
```

2. Abstract Interfaces

This Service provides one functionality.

2.1. POST:

This is used create a new record to add the current thermostat set-point value to the thermostat device manager using POST API.

3. Abstract Information Model

The output of the service is the current SetPoint value of the Thermostat.

Table 1 SetPoint Thermo Service Request

Field	Description
serviceType	Get the current SetPoint of the thermostat and create a record into the thermostat device manager and store the value with time stamp
deviceType	Thermostat

Table 2 SetPoint Thermo Service Response

Field	Description
Output	It gives the value of thermostat setpoint

4. Non-functional Requirements

The service has none non-functional requirements.

5. Revision history

5.1. Amendments

No.	Date	Version	Subject of Amendments	Author
1	2021-01-6	0.1	First draft	Salman Javed

5.2. Quality Assurance

No.	Date	Version	Approved by
1			